Daphnia magna nutrition across cyanobacterial and eukaryotic algal diets

When fed increasing proportions of cyanobacteria (*Microcystis aeruginosa* or *Synechococcus elongatus*), *Daphnia magna* suffers progressive declines in survival, growth, and reproduction driven by the combined absence of dietary sterols and long-chain polyunsaturated fatty acids (PUFAs), with toxic *Microcystis* imposing additional microcystin-mediated and protease-inhibitor-mediated harm. The freshwater eustigmatophyte *Nannochloropsis limnetica* serves as a high-quality food because it uniquely supplies both cholesterol and eicosapentaenoic acid (EPA), the two nutrients most limiting for Daphnia on cyanobacterial diets. (PubMed Central +2) Sterol limitation constrains somatic growth first; once sterol needs are met, EPA becomes the next bottleneck, (PubMed Central) primarily limiting reproduction. (Springer +5) These two deficiencies interact synergistically rather than following Liebig's law of the minimum (nih) — meaning that co-limited Daphnia perform worse than single-nutrient models predict. (Semantic Scholar +2) Mixed-diet experiments reveal that **at least ~50% eukaryotic (sterol- and PUFA-containing) algal carbon** is required in the diet to sustain adequate growth and reproduction, (Springer) (Oxford Academic) and that even 10–20% toxic *Microcystis* measurably depresses fitness.

Nutritional profiles of the three algae diverge in ways that matter most

The three algae in this system differ radically in their content of the two nutrient classes most limiting for Daphnia: sterols and long-chain PUFAs.

Microcystis aeruginosa, like all cyanobacteria, **completely lacks sterols** (PubMed Central) and contains **no EPA, DHA, or ALA** (PubMed Central) (Ahlgren et al. 1992; Von Elert et al. 2003; Bec et al. 2006). Its fatty acid profile is dominated by saturated and monounsaturated C16 chains. (DNB) Beyond these biochemical deficiencies, toxic strains produce **microcystins** (cyclic heptapeptide hepatotoxins; (ScienceDirect) >250 structural variants, with MC-LR most common and most toxic), (Wikipedia) **cyanopeptolins**, **aeruginosins**, and other protease inhibitors that directly impair Daphnia digestive physiology. (MDPI +2) Intracellular concentrations of chymotrypsin inhibitors CP954 and BN920 in *M. aeruginosa* PCC 7806 reach **1,550 and 120 μmol L^{-1}**, respectively (Von Elert et al. 2012). (PubMed) Colony-forming strains additionally interfere mechanically with the filtering apparatus. Microcystin content in PCC 7806 has been measured at **4.8–5.6 fg cell^{-1}** (PubMed) (Sadler & Von Elert 2014).

Synechococcus elongatus (strain SAG 89.79, the standard experimental isolate) is a non-toxic picocyanobacterium (Frontiers) (~1 μm) that serves as a model of pure biochemical deficiency. (DNB) It **completely lacks both sterols and PUFAs**, (The Free Library +3) containing only palmitic (16:0) and palmitoleic (16:1n-7) acids (Wiley +2) (DeMott & Müller-Navarra 1997; Bec et al. 2006; Santos-Merino et al. 2018). It encodes only a single Δ9-desaturase, making it incapable of producing any polyunsaturated fatty acid. (PubMed Central) Its small cell size is within the filtration range of *D. magna* (mesh size 0.4–0.7 μm; Brendelberger 1991), (PubMed) so poor performance is not a feeding-efficiency problem but a nutritional one. Because it is non-toxic and unicellular, it cleanly isolates the sterol + PUFA deficiency from toxicity and morphological interference. (Frontiers +3)

Nannochloropsis limnetica (SAG 18.99, a freshwater eustigmatophyte) stands in sharp contrast. It is **rich in EPA** (d-nb) — typically **24% of total fatty acids**, reaching **55.56 mg g^{-1} dry weight** under high-phosphate

culture conditions (Krienitz & Wirth 2006) ScienceDirect — and its sterol profile is **dominated by cholesterol** d-nb ResearchGate (>75–80% of total sterols), the directly usable animal sterol (Volkman et al. 1992; Martin-Creuzburg et al. 2009, 2015). It also contains arachidonic acid (**10.52 mg g^{-1} DW**) and linoleic acid (**22.19 mg g^{-1} DW**). ScienceDirect It does not produce DHA, but Daphnia primarily require EPA rather than DHA. PubMed Central Its cell size (2–4 μm) is readily ingestible. The fatty acid and sterol profiles of *N. limnetica* are virtually identical to marine congeners like *N. oculata* and *N. gaditana* (Freire et al. 2016), confirming that data from marine Nannochloropsis species are transferable. ScienceDirect

Nutrient	*M. aeruginosa*	*S. elongatus*	*N. limnetica*
EPA (20:5ω3)	Absent	Absent	~24% of TFA; up to 55.6 mg g^{-1} DW
DHA (22:6ω3)	Absent	Absent	Absent
ALA (18:3ω3)	Trace or absent	Absent	Present (intermediate)
ARA (20:4ω6)	Absent	Absent	~10.5 mg g^{-1} DW
Sterols	None detected	None detected	Cholesterol-dominant (>75%)
Toxins	Microcystins, protease inhibitors	None	None
Digestibility	Reduced by PIs + colony morphology	Good (well-assimilated)	Good

Sole-diet performance reveals a hierarchy from lethal to excellent

Microcystis aeruginosa: lethal to severely debilitating

A 100% toxic *M. aeruginosa* diet is **lethal or near-lethal**, Wiley Online Library with median survival measured in days to weeks depending on clone and strain (Rohrlack et al. 2001; Lürling 2003). Even non-toxic *Microcystis* supports neither growth nor reproduction long-term due to sterol and PUFA absence. PubMed Central +3 The 48-hour LC_{50} for cell-bound microcystins in *D. magna* is approximately **85.72 μg L^{-1}** (among the lowest recorded for cladocerans), PubMed while dissolved MC-LR has an EC_{50} of **~13.5 mg L^{-1}** ScienceDirect — orders of magnitude higher, underscoring that cell-bound exposure via ingestion is the primary toxicity route. Reproductive effects appear at concentrations as low as **2.5 μg L^{-1}** cell-bound microcystins. PubMed

On toxic *Microcystis*, Daphnia exhibit dramatically reduced somatic growth, delayed maturity, smaller clutches of smaller neonates, and often negative population growth rates. MDPI ACS Publications A compensatory reproductive strategy has been documented: cyanobacteria-fed mothers invest in **more but smaller eggs** (r-strategy), PubMed Central allocating relatively more energy to reproduction as a risk-spreading mechanism

PubMed PubMed Central (Bednarska et al. 2022; Zhou et al. 2020). Toxic *Microcystis* can also trigger production of resting eggs (ephippia) via induction of sexual reproduction. ScienceDirect

Synechococcus elongatus: the clean model of nutritional inadequacy

D. magna* cannot survive long-term on pure *S. elongatus The Free Library +3 (Bec et al. 2006; Martin-Creuzburg et al. 2005). Growth rates are near-zero or negative after maternal lipid reserves are depleted within 2–3 days (DeMott & Müller-Navarra 1997). Semantic Scholar No eggs are produced on a pure *Synechococcus* diet. DNB Supplementation experiments have definitively separated the contributions of each deficiency: **cholesterol supplementation alone significantly rescues somatic growth**; Springer +2 **EPA supplementation alone does not improve growth** unless sterols are already adequate; Springer **combined cholesterol + EPA supplementation produces the greatest improvement**, Royal Society Open Science approaching (but not fully reaching) growth rates on eukaryotic algae ResearchGate +2 (Von Elert & Wolffrom 2001; Martin-Creuzburg et al. 2008, 2009). Egg production, by contrast, is primarily limited by PUFA absence Academia.edu Springer (Martin-Creuzburg & Von Elert 2009). Springer

Nannochloropsis limnetica: a high-quality reference food

N. limnetica is used as a **positive control** in Daphnia food-quality research d-nb by the Martin-Creuzburg, Wacker, and Von Elert laboratories. It supports high somatic and population growth rates, normal clutch sizes, and high survival. Wiley When *D. magna* is fed *N. limnetica*, eggs contain **2.3-fold higher n-3 PUFA concentrations** than somatic tissues (Wacker & Martin-Creuzburg 2007). Academia.edu The cholesterol- and EPA-rich diet eliminates genotype differences in lifespan observed on PUFA-poor diets Springer (Yampolsky et al. 2023). ResearchGate Because *N. limnetica* simultaneously satisfies both the sterol and PUFA requirements, it does not impose either form of nutritional limitation — a rare combination among freshwater microalgae.

Mixed diets reveal dose-dependent declines and interactive co-limitation

The sterol-PUFA co-limitation framework

The most important conceptual advance in this literature is the **interactive co-limitation model** of Martin-Creuzburg, Sperfeld, and Wacker (2009). When *D. magna* is fed *S. elongatus* supplemented with increasing cholesterol ± saturating EPA — or, equivalently, increasing proportions of *N. limnetica* blended with *S. elongatus* — growth follows a hierarchical pattern: Royal Society Open Science sterol limitation constrains growth first; once sterols reach saturation Royal Society Open Science (~8.9 μg cholesterol mg C^{-1} for *D. magna*; Martin-Creuzburg et al. 2014), EPA becomes limiting. Royal Society Open Science Critically, Sperfeld et al. (2012, *Ecology Letters*) showed that **Liebig's law of the minimum consistently overestimates growth at co-limiting conditions**. The interactive multiple-limitation hypothesis (MLH) fits the data significantly better, Connect H1 implying that sterols and EPA act synergistically: having adequate levels of one nutrient reduces the threshold requirement for the other. Connect H1

Maximum growth is achieved at approximately **≥13.7 μg cholesterol mg C^{-1} and ≥2.8 μg EPA mg C^{-1}** Academia.edu (Sperfeld et al. 2012). These thresholds shift with temperature — cholesterol demand increases at higher temperatures The Company of Biologists (Sperfeld & Wacker 2009), The Company of Biologists while EPA

demand increases at lower temperatures (bioRxiv) (PeerJ) (Sperfeld & Wacker 2012) (PubMed Central) (Springer) — a pattern consistent with homeoviscous adaptation of membrane lipids.

Non-toxic cyanobacteria: gradual, approximately linear decline

When *Scenedesmus obliquus* (sterol-containing) is mixed with *S. elongatus* (sterol-free) at ratios of 100:0 through 0:100, somatic growth, population growth, maternal dry mass, offspring number, and survival all decline (Springer) **approximately linearly** with increasing cyanobacterial proportion (Springer) (PubMed) (Martin-Creuzburg et al. 2005). (Springer) **At least ~50% eukaryotic algal carbon** is required to avoid sterol deficiency. (Oxford Academic) *D. galeata* has a twofold higher incipient limiting sterol level than *D. magna*, making it more sensitive to sterol dilution. (PubMed) (Springer)

The key three-species gradient study (Sarrazin & Sperfeld 2022) used mixtures of *Acutodesmus obliquus* (ACU), *N. limnetica* (NAN), and *S. elongatus* (SYN) at compositions ranging from 5:40:55 (good quality) through 5:20:75 (medium) to 5:5:90 (poor). Somatic growth rate decreased with increasing SYN proportion across all six *D. magna* clones tested. (d-nb) Notably, at 90% SYN, daphnids achieved **surprisingly high short-term growth rates** — attributed to maternal investment from mothers raised on high-quality diets (DNB) (60:40 ACU:NAN), demonstrating significant transgenerational buffering of nutritional stress.

Toxic Microcystis: sharper thresholds at lower proportions

Toxic *M. aeruginosa* imposes dose-dependent declines that are steeper than those from non-toxic cyanobacteria, because toxicity, protease inhibition, and feeding interference compound the nutritional deficiency. Studies using mixtures of *M. aeruginosa* with *Chlorella* or *Scenedesmus* reveal:

- **10–20% toxic *Microcystis***: Significant negative effects on survival, development, and reproduction (ScienceDirect) (multiple studies reviewed in Schwarzenberger et al. 2013).
- **20–35%**: Reduced population carrying capacity and suppressed growth at both 20% and 35% *Microcystis* in mixture with *Chlorella pyrenoidosa*. (Nature) (Science.gov)
- **>60% (with protease inhibitors)**: Growth rates significantly reduced beyond the capacity of inducible tolerance mechanisms (Von Elert et al. 2012). (PubMed)
- **75–100%**: Feeding essentially ceases; gut transit time drops from ~27 minutes (pure *Scenedesmus*) to ~20 minutes at 75% cyanobacteria; at 100% toxic *Microcystis*, ingestion is insignificant. (ResearchGate)

A critical finding by Lürling (2003) is that even **non-toxic *Microcystis*** mixed with *Scenedesmus* significantly reduces *D. magna* growth — attributable to feeding inhibition by non-microcystin metabolites and mechanical interference, not toxins alone. (PubMed) This means the dose-response relationship for *Microcystis* reflects multiple overlapping mechanisms: nutritional deficiency, protease inhibition, microcystin toxicity, and physical feeding interference.

PUFA dynamics in Daphnia tissues track diet rapidly but with selective retention

Daphnia body fatty acid composition **strongly reflects diet** (Frontiers) (PubMed Central) (Brett et al. 2006), with major compositional shifts occurring within **~4 days** of a diet switch (Google Scholar) (Taipale et al. 2011).

However, essential C20 PUFAs (EPA and ARA) are **selectively retained for >14 days** after switching to a PUFA-poor diet — evidence of physiological prioritization. Compound-specific ^{13}C labeling has directly demonstrated **elongation and desaturation of ALA (18:3ω3) to EPA** in *D. magna* (Frontiers) approximately 10 days after switching to an EPA-deficient diet (Google Scholar) (Taipale et al. 2011), confirming that Daphnia possess the enzymatic machinery for C18→C20 conversion. Schlechtriem et al. (2006) documented this conversion in *D. pulex* as well, with enhanced conversion at colder temperatures (ResearchGate) (11°C vs. 22°C). (PubMed) **However, the conversion rate is insufficient to meet full physiological demand** — Daphnia remain functionally dependent on dietary EPA. (The Company of Biologists) (PeerJ)

On *N. limnetica* diets, Daphnia preferentially allocate EPA to eggs over somatic tissue. (Frontiers) Eggs contain **2.3-fold higher n-3 PUFA concentrations** than soma (Wacker & Martin-Creuzburg 2007). (Academia.edu) At 15°C, females allocate **twice as much EPA to offspring** as at 20°C, consistent with the greater membrane fluidity demands of cold environments (PeerJ) (Sperfeld & Wacker 2012). (ResearchGate) Neonate EPA content is approximately **0.44 ng μg^{-1} dry weight**, with total PUFA at **~18.3 ng μg^{-1} dry weight** (Springer) (Sikora et al. 2016). When *D. magna* switches from *Scenedesmus* to *Microcystis*, PUFA and carbon turnover decrease significantly, and animals prioritize PUFA allocation to somatic growth over reproduction — an adaptive strategy for population maintenance during food quality deterioration. (Frontiers)

The EPA threshold for 75% growth saturation in *D. magna* at 20°C ranges from **0.7 to 7.3 μg EPA mg C^{-1}** across studies (Sperfeld & Wacker 2011; Sikora et al. 2016; Ilić et al. 2019), (Wiley Online Library) with variation likely reflecting differences in sterol adequacy and experimental conditions. (Springer) Importantly, Ilić et al. (2019) showed that **EPA and ARA are functionally substitutable** for somatic growth — dose-response curves for EPA, ARA, and 50:50 mixtures were "virtually congruent" (MDPI) (ResearchGate) — though EPA has additional roles in eicosanoid signaling (Fink & Windisch 2019).

Sterol biology constrains the cyanobacteria–Daphnia interface

Daphnia are **obligate sterol auxotrophs** — they cannot synthesize sterols de novo and must obtain them from diet (Springer +2) (Goad 1981; (PubMed Central) Martin-Creuzburg et al. 2014). Cholesterol is the predominant body sterol and serves as the precursor for ecdysteroid molting hormones, (Oxford Academic) a structural membrane component, and a signaling molecule. (Frontiers) (The Company of Biologists) Ten different phytosterols have been tested for their capacity to support *D. magna* growth, (Semantic Scholar) (Springer) with **threshold concentrations ranging from 3.5 to 34.4 μg mg C^{-1}** (Martin-Creuzburg et al. 2014). (Wiley Online Library) Δ5-sterols (sitosterol, stigmasterol) and Δ5,7-sterols (ergosterol, 7-dehydrocholesterol) adequately support growth; Δ7-sterols (Semantic Scholar) (common in Chlorophyte green algae like *Scenedesmus*) are less efficiently converted. (Springer +2) Daphnia preferentially take up cholesterol over phytosterols requiring bioconversion (Wiley Online Library) (Laine et al. 2024). (Ovid)

This is why **cyanobacteria are such poor food**: they are completely sterol-free. (ResearchGate +3) Martin-Creuzburg et al. (2008) demonstrated across multiple cyanobacterial genera (*Synechococcus*, *Anabaena*, *Aphanizomenon*, *Microcystis*) that somatic growth is **"primarily constrained by the absence of sterols, provided that the cyanobacterial carbon is readily ingested and non-toxic."** (Academia.edu +5) And it is precisely why *N. limnetica* is such effective food: its cholesterol is directly usable without bioconversion, simultaneously satisfying the most critical nutritional constraint.

Body cholesterol content increases with dietary supply and is **enhanced at higher temperatures** (The Company of Biologists +2) (Sperfeld & Wacker 2009). (PubMed) (The Company of Biologists) The **female sterol-limited growth threshold (~9.6 μg mg C^{-1}) is higher than the male threshold (~6.2–6.4 μg mg C^{-1})**, making females more susceptible to sterol deficiency (Frontiers) (Martin-Creuzburg et al. 2018). (Frontiers) When EPA is present, the sterol threshold is reduced to ~8.2 μg mg C^{-1} in females (Frontiers) — further evidence of interactive co-limitation.

Microcystin toxicology: body burden, detoxification, and the limits of tolerance

Microcystins inhibit **protein phosphatases PP1 and PP2A**, disrupting phosphorylation-dependent signaling, and induce oxidative stress. (ScienceDirect) Body burden increases with exposure concentration and duration. (Science.gov) Using high-resolution LC-MS, Sadler & Von Elert (2014) found unconjugated MC-LR in Daphnia tissues fed toxic *M. aeruginosa* PCC 7806 but detected **no glutathione-MC-LR conjugates**, contradicting the long-assumed GST-mediated detoxification pathway identified *in vitro* by Pflugmacher et al. (1998). GST activity increased identically when Daphnia fed on microcystin-producing wild-type or microcystin-free mutant PCC 7806, suggesting the GST response is to cyanobacteria generally rather than microcystins specifically. (PubMed) (Science.gov)

Instead, the primary tolerance mechanisms involve:

- **Digestive protease remodeling**: When exposed to chymotrypsin inhibitors from *Microcystis*, *D. magna* replaces inhibitor-sensitive gut chymotrypsins with **insensitive isoforms** within its lifetime — an inducible phenotypic plastic response (PubMed Central) (Von Elert et al. 2012). (PubMed) This tolerance is **maternally transferred**: offspring from exposed mothers show higher growth rates on protease-inhibitor diets (Schwarzenberger & Von Elert 2013).
- **Transporter gene regulation**: RNAseq identified membrane transporter genes (including ABC transporters) as specifically regulated by microcystins, likely functioning to export toxins from cells (Schwarzenberger et al. 2014). Expression differences correlate with clonal tolerance levels. (PubMed)
- **Gut microbiome mediation**: In a landmark finding, Macke et al. (2017, *Nature Communications*) showed that interclonal variation in *Microcystis* tolerance **disappeared** when Daphnia were made germ-free and inoculated with identical microbiota, and instead mirrored the microbiota donor's tolerance level. (Nature) Flavobacteriia were identified as candidates for conferring high tolerance. (Wiley Online Library)
- **Microevolutionary adaptation**: Populations from lakes with frequent cyanobacterial blooms show higher IC_{50} values for their digestive chymotrypsins, reduced chymotrypsin gene copy numbers, and positive selection on protease gene sequences (MDPI) (ScienceDirect) (Schwarzenberger et al. 2017, 2020, 2021).

At the population level, the **>60% *Microcystis* threshold** identified by Von Elert et al. (2012) represents the boundary beyond which inducible protease remodeling can no longer compensate for protease inhibitor load. (PubMed) Below this threshold, tolerant *D. magna* clones maintain near-normal growth; above it, even tolerant clones are significantly impaired.

Complementarity, compensation, and selective feeding in mixed food environments

The concept of **nutritional complementarity** was first demonstrated for Daphnia by DeMott (1998), who showed that two individually poor foods — P-rich but PUFA-deficient *Synechococcus* and P-deficient but PUFA-rich *Scenedesmus* — produced improved growth in mixtures through complementary resource supply. The Free Library This principle extends to the sterol-PUFA system: mixing sterol-free *S. elongatus* with sterol- and EPA-rich *N. limnetica* naturally releases co-limitation in a dose-dependent manner, PubMed Royal Society Open Science with growth saturating at ~50% *N. limnetica* (Martin-Creuzburg et al. 2009). Royal Society Open Science

D. magna exhibits limited selective feeding capacity. Using video recording and Ivlev's Electivity Index, Ghadouani et al. (2004) showed that Daphnia preferentially ingest green algae over cyanobacteria in mixed suspensions — but **this selectivity is lost when cyanobacteria dominate (>75%)** the food environment. ResearchGate Hunger overrides rejection: after 24 hours of food deprivation, previously rejecting Daphnia resume feeding on toxic cells (DeMott 1993). MDPI Compensatory feeding (increased filtration rate on poor food) occurs on some low-quality monocultures but is **suppressed by *Microcystis***, whose metabolites inhibit feeding activity even independent of microcystins Taylor & Francis Online PubMed (Rohrlack et al. 1999, 2001). Morphological plasticity of the filtering apparatus — including decreased mesh size and enlarged filter screen area — has been documented in response to cyanobacterial exposure PubMed Central (Bednarska & Dawidowicz 2007; Ghadouani & Pinel-Alloul 2002).

A further compensatory pathway operates through the microbial food web: heterotrophic protists feeding on picocyanobacteria can **trophically upgrade** this carbon for Daphnia. Inter Research The nanoflagellate *Paraphysomonas* synthesizes PUFAs and sterols de novo when feeding on *S. elongatus*, significantly enhancing *D. magna* growth; DNB ciliates (*Colpidium*, *Cyclidium*) produce tetrahymanol and hopanoids that serve as functional sterol substitutes d-nb +2 (Martin-Creuzburg et al. 2005b; Bec et al. 2006).

Conclusion

This body of work, developed largely by the Martin-Creuzburg, Von Elert, Wacker, and Sperfeld research groups over two decades, establishes that cyanobacterial dominance in Daphnia diets creates a multi-dimensional nutritional crisis. **Sterol absence is the primary somatic growth constraint; PUFA (EPA) absence is the primary reproductive constraint; and these interact synergistically rather than additively.** Academia.edu +4 *N. limnetica* occupies a uniquely favorable niche as a freshwater alga providing both cholesterol and EPA, making it an ideal experimental counterpart to sterol- and PUFA-free *S. elongatus*. Toxic *Microcystis* adds microcystin toxicity and protease inhibition on top of the shared cyanobacterial nutritional deficiencies, ScienceDirect Google Scholar creating sharper dose-response thresholds (~10–20% measurable effects, >60% overwhelming tolerance). Daphnia possess multiple compensatory mechanisms — limited PUFA bioconversion (C18→C20), Google Scholar PubMed Central inducible protease remodeling, maternal lipid provisioning, Frontiers gut microbiome-mediated tolerance, and selective feeding — but none fully compensates for high cyanobacterial dietary proportions. PubMed Central PeerJ The interactive co-limitation model, rather than Liebig's law, best predicts Daphnia performance across dietary gradients, Semantic Scholar +2 with important temperature-dependent modulation of both sterol and EPA requirements. PubMed Central +2 These findings have

direct ecological relevance as climate warming simultaneously increases cyanobacterial bloom frequency and alters the stoichiometric and biochemical demands of zooplankton grazers. d-nb +2